

Outcome Based Education

OBE Implementation Phases

- **Phase I**

- Framing Curriculum
- Planning Cocurricular activities
- Planning Extracurricular activities

- **Phase II**

- Content Delivery
- Conducting assessments
- Organising Cocurricular and Extracurricular events

- **Phase III**

- Computing attainments
- Gap Analysis
- Taking remedial measures

Program Outcomes (POs)

PROGRAM OUTCOMES (POs)	
PO1	Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	Problem analysis : Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3	Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO4	Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5	Modern tool usage : Create, select and apply appropriate techniques, resources and modern engineering and IT tools, including prediction and modelling to complex engineering activities, with an understanding of the limitations.
PO6	The engineer and society : Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice
PO7	Environment and sustainability : Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO8	Ethics : Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO9	Individual and team work : Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings
PO10	Communication : Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	Project management and finance : Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	Life-long learning : Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcomes (PSOs)

Mechanical Engineering

- **PSO-1:** Identify, analyze and impart complex engineering knowledge in Thermal Engineering, Design Engineering and Manufacturing Engineering domains.
- **PSO-2:** Enabling the students to take-up career in core industries so as to develop products using CAD/CAM tools.

The Outcomes (POs, PSOs)

**The POs, PSOs need to be addressed through the
Course Outcomes (COs)
of Existing Courses
of the Curriculum**

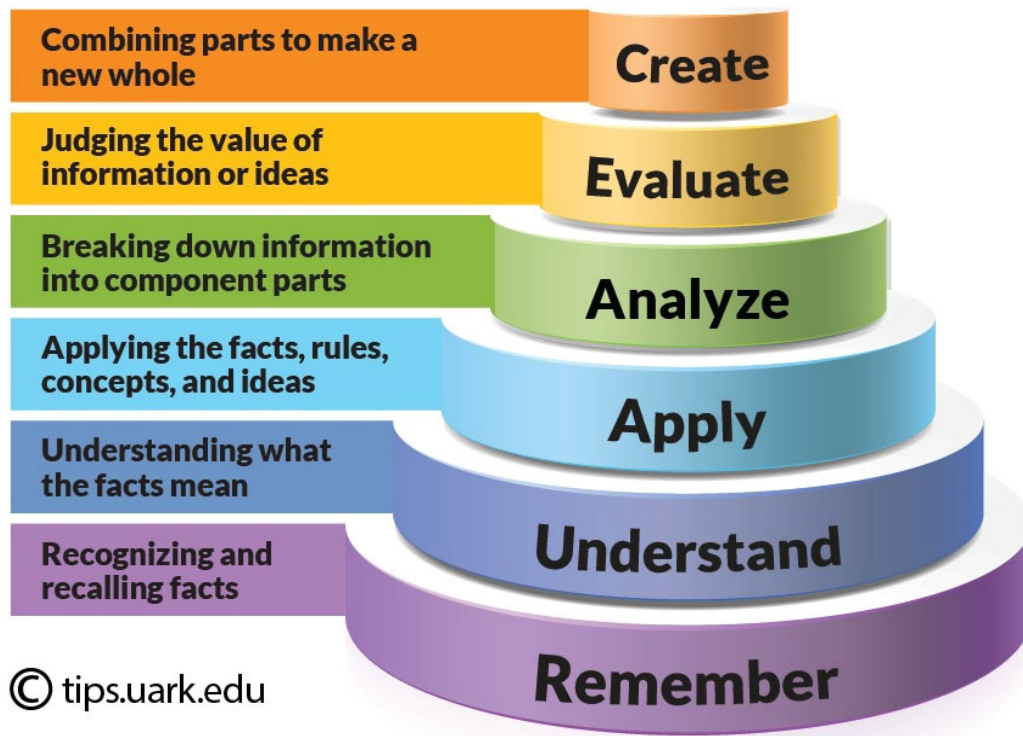
Characteristics of COs

- Describe skills/competencies gained by student **at the end of the course**
- Every CO shall have **action verb(s)**
- Need to be **assessable** and **measurable**
- Every course has a set of COs
- A course may have about **6 COs**
- Every CO addresses **one or more outcomes** (POs/PSOs)
- Are aligned with one or more POs
- Are closely related to the assessments

OBE and Bloom's Taxonomy

- **Bloom's taxonomy** is a set of hierarchical models used to classify educational learning objectives into levels of complexity and specificity.
- **Bloom's taxonomy** is a system used to define and distinguish different levels of human cognition—i.e., thinking, learning, and understanding.

Different Cognitive Level



Cognitive Level	Notation Used
Create	K6
Evaluate	K5
Analyze	K4
Apply	K3
Understand	K2
Remember	K1

Remembering

The learner is able to recall, restate and remember learned information

- Describing
- Finding
- Identifying
- Listing



- Retrieving
- Naming
- Locating
- Recognizing

Can students recall information?

Understanding

Student grasps meaning of information by interpreting and translating what has been learned

- Classifying
- Comparing
- Exemplifying
- Explaining



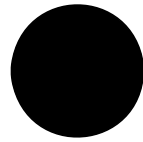
- Inferring
- Interpreting
- Paraphrasing
- Summarizing

Can students explain ideas or concepts?

Applying

Student makes use of information in a context different from the one in which it was learned

- Implementing
- Carrying out



=



- Using
- Executing

Can students use the information in another familiar situation?

Analyzing

Student breaks learned information into its parts to best understand that information

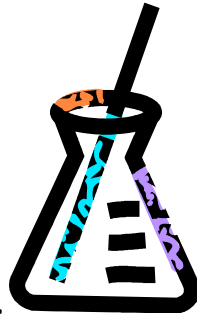
- Attributing
 - Comparing
 - Deconstructing
 - Finding
- 
- Integrating
 - Organizing
 - Outlining
 - Structuring

Can students break information into parts to explore understandings and relationships?

Evaluating

Student makes decisions based on in-depth reflection, criticism and assessment

- Checking
- Critiquing
- Detecting
- Experimenting



- Hypothesising
- Judging
- Monitoring
- Testing

Can students justify a decision or a course of action?

Creating

Student creates new ideas and information using what previously has been learned

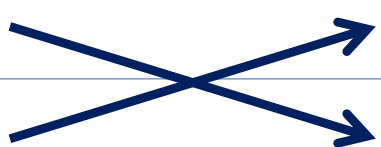
- Constructing
- Designing
- Devising
- Inventing



- Making
- Planning
- Producing

Can students generate new products, ideas, or ways of viewing things?

A comparison

Bloom's Levels		Program Outcomes (POs)	
Remember (K1)			
Understand (K2)			
Apply (K3)		Apply Knowledge (PO1)	
Analyze (K4)		Problem Analysis (PO2)	
Evaluate (K5)		Design/Development of Solutions (PO3)	
Create (K6)		Conduct Investigations (PO4)	

Program Outcomes (POs)

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Developing the Remember Attribute

Course Outcome: Example -A1

At the end of the course the student will have the ability to **explain** the importance of Fourier Transform

Course Outcome: Example -A1 Assessment

- **State and explain** Fourier Transform
- **State and explain** the applications of Fourier Transform
- **Write Short Notes** on Fourier Transforms

Course Outcome: Example -A2

At the end of the course the student will have the ability to **describe** the working of Operational Amplifiers

Course Outcome: Example -A2 Assessment

- **Explain** with a Block Diagram the working of an Operational Amplifier (OPAMP)
- **List** the various stages of an OPAMP and discuss their functioning
- **Write Short Notes** on Frequency Compensation

Attribute Developed

REMEMBER

QUIZ Corner!

- Is the **REMEMBER** attribute emphasized by NBA?
- Do our question papers (assessments) contain such type of questions?
- Do we need to plan and develop this skill?

The Program Outcomes (POs)

PO1 Apply Knowledge

- PO2 Problem Analysis**
- PO3 Design/Development of Solution**
- PO4 Conduct Investigations**
- PO5 Use Modern Tool**
- PO6 Engineer and Society**
- PO7 Environment and Sustainability**
- PO8 Professional Ethics**
- PO9 Individual and Team work**
- PO10 Communicate Effectively**
- PO11 Project Management and Finance**
- PO12 Life-Long Learning**

PO-1: Apply Knowledge

Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialisation to the solution of complex engineering problems.

Course Outcome: Example -B1

CO2: At the end of the course the student will have the ability to **obtain** the Fourier Transform of a given signal

Course Outcome: Example -B1

Assessment -1

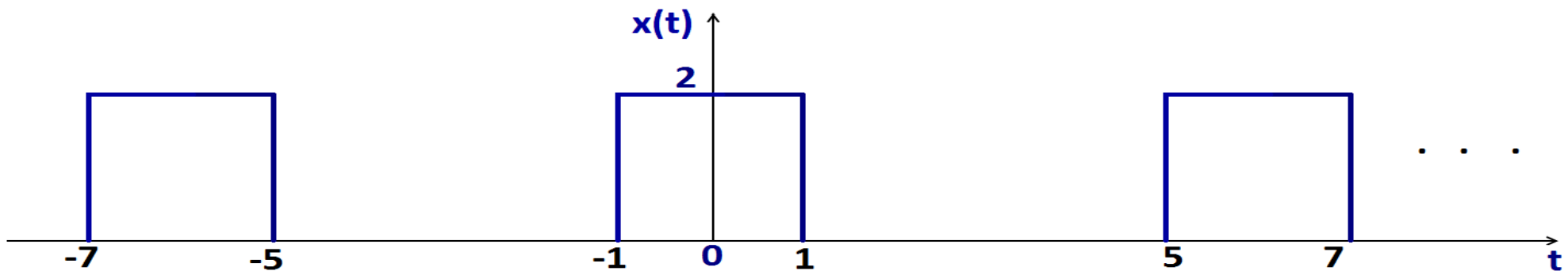
Obtain the of Fourier Transform of the signal given below:

$$x(t) = e^{-4t} u(t)$$

Course Outcome: Example -B1

Assessment -2

Obtain the of Fourier Transform of the signal given below:



Course Outcome: Example –B2

CO2: At the end of the course the student will have the ability to **apply knowledge of mathematics, physics and electronics** to obtain the transfer function of the given Analog Electronic Circuit

Course Outcome: Example –B2

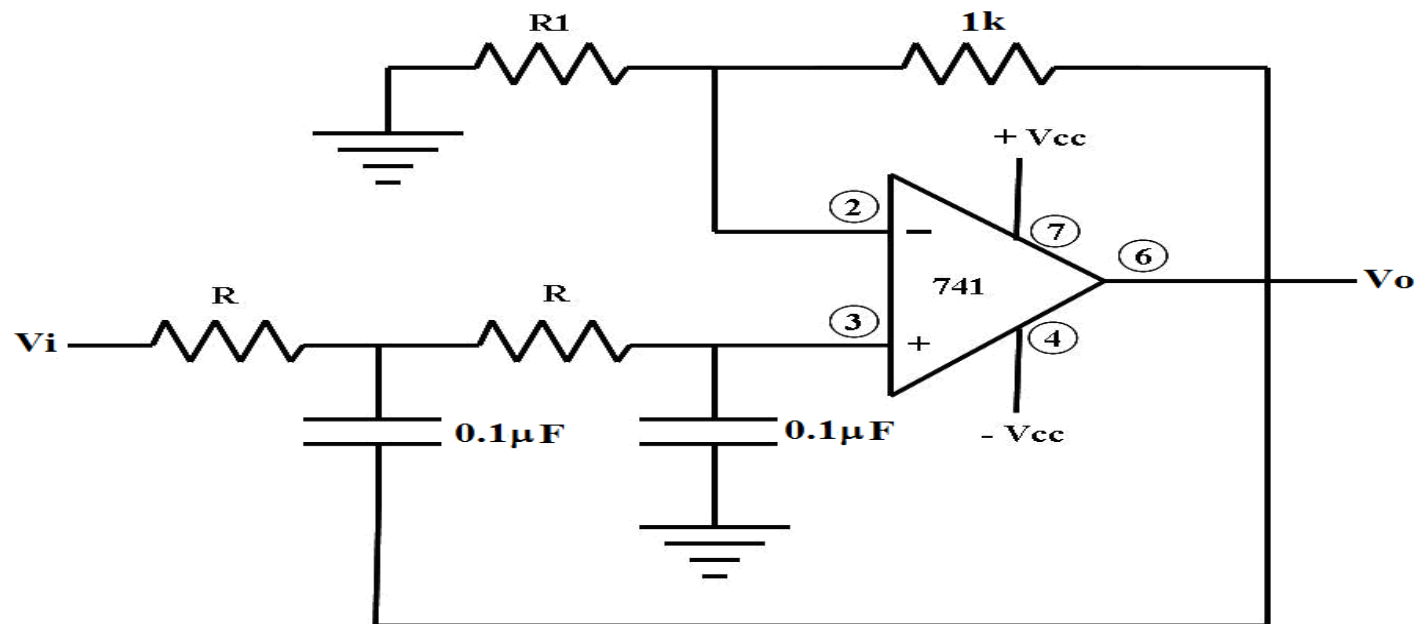
Assessment -1

Obtain the of Transfer Function of the First Order Active LPF realized using the Operational Amplifier

Course Outcome: Example –B2

Assessment -2

Obtain the Transfer Function of the circuit given below



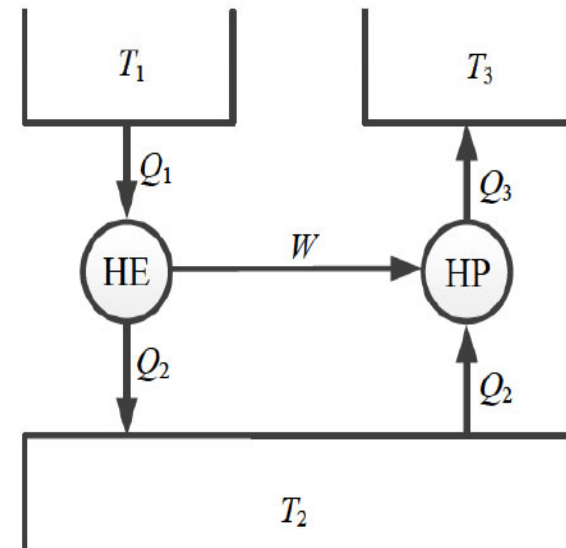
Course Outcome: Example –B3

Course outcome of Thermodynamics

- **CO2: Apply** the laws of thermodynamics to closed and open engineering systems and calculate the property changes and performance.

Example for CO2

The figure shows a heat engine (HE) working between two reservoirs. The amount of heat (Q_2) rejected by the heat engine is drawn by a heat pump (HP). The heat pump receives the entire work output (W) of the heat engine. If temperatures, $T_1 > T_3 > T_2$, then the relation between the efficiency (η) of the heat engine and the coefficient of performance (COP) of the heat pump is



(A) $\text{COP} = \eta$

(B) $\text{COP} = 1 + \eta$

(C) $\text{COP} = \eta^{-1}$

(D) $\text{COP} = \eta^{-1} - 1$

Course Outcome: Example –B4

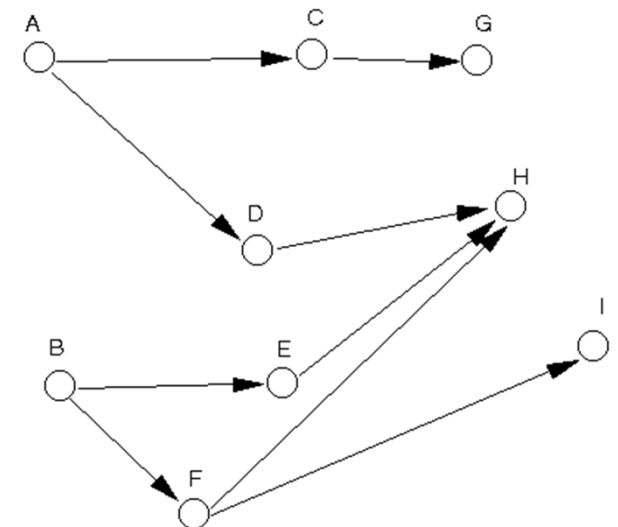
Course Name	Course Outcome
Data Structures and Algorithms	Apply appropriate operations on linear/non-linear data structure for solving a given problem.

Example: Implement the Quick Sort algorithm for the given array : 5, 8, 44, 3, 12, 45, 16, 20. Provide the recurrence equation and derive its time complexity in both average case and worst case.

Course Outcome: Example –B4

Course Name	Course Outcome
Network Security	????

Example: Examine the uncertainty completion time for the nodes B-E-H.



The Program Outcomes (POs)

- PO1 Apply Knowledge
- PO2 Problem Analysis**
- PO3 Design/Development of Solution
- PO4 Conduct Investigations
- PO5 Use Modern Tool
- PO6 Engineer and Society
- PO7 Environment and Sustainability
- PO8 Professional Ethics
- PO9 Individual and Team work
- PO10 Communicate Effectively
- PO11 Project Management and Finance
- PO12 Life-Long Learning

PO-2: Problem Analysis

*Identify, formulate, research literature, and **analyse** complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences*

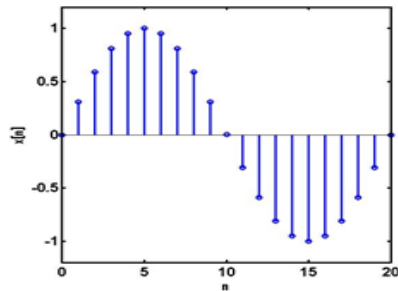
Course Outcome: Example -C1

CO3: At the end of the course the student will be able to **analyze** the Fourier Transform Pair of an LTI system and arrive at suitable conclusions

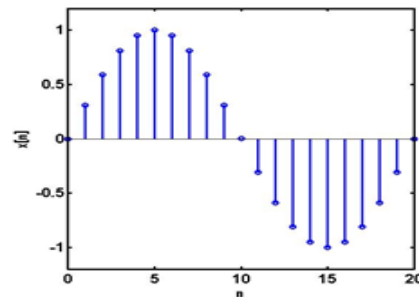
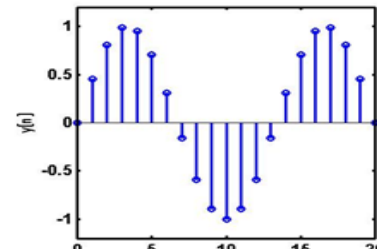
Course Outcome: Example -C1

Assessment -1

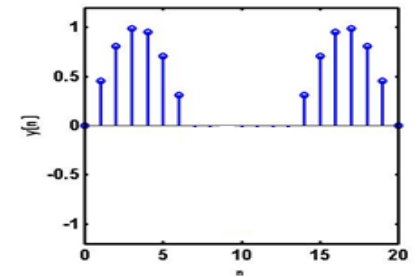
Analyze the following systems for linearity



*Discrete Time
System*



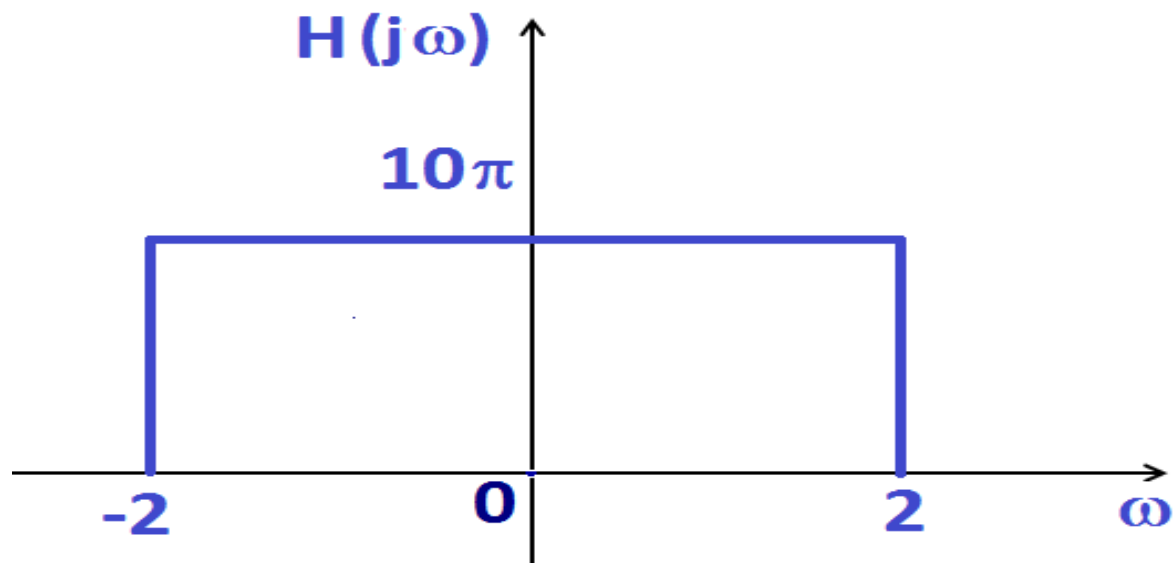
*Discrete Time
System*



Course Outcome: Example -C1

Assessment -2

Analyze the system given below for Stability and Realizability



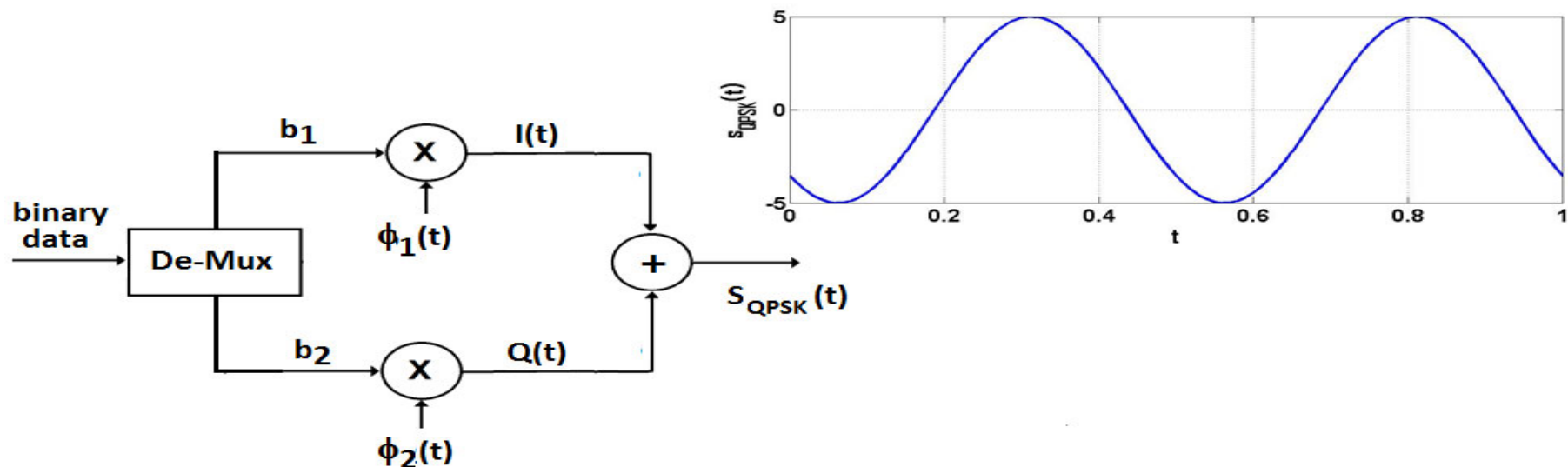
Course Outcome: Example –C2

CO3: At the end of the course the student will be able to **analyze** the digital communication system and arrive at suitable conclusions

Course Outcome: Example –C2

Assessment -1

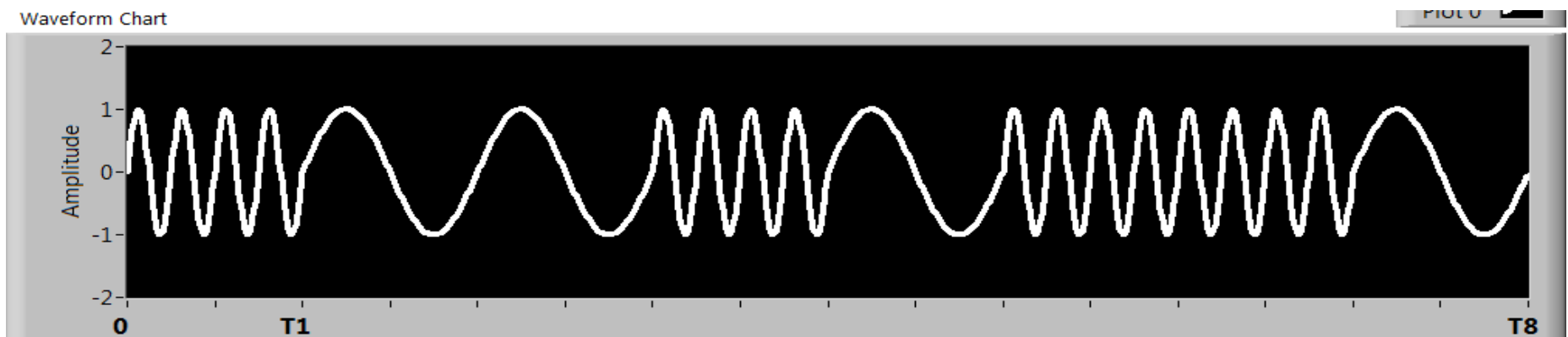
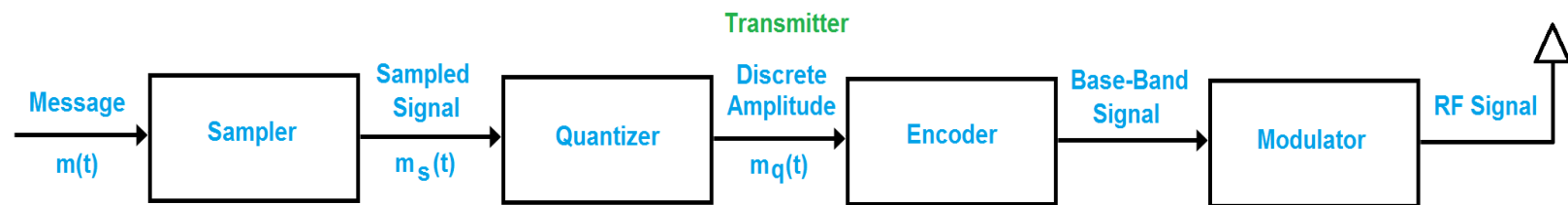
Analyze the Block Diagram given below to obtain the modulation scheme and hence the input binary bit sequence



Course Outcome: Example –C2

Assessment -2

Analyze the information given below to obtain the modulation scheme, the binary sequence and the input sample value (Assume EO standard)



Course Outcome: Example –C3

Course Name	Course Outcome
Design and Analysis of Algorithm	Analyze and choose appropriate problem solving strategy to solve problems.

Assessment:

Set of Items with weights and value with $W=12$:

Items[i] →	1	2	3	4
val[i]	25	15	20	36
wt[i]	7	2	3	6

Infer the solution for the given problem by identifying appropriate algorithm and list the recurrence equations.

Course Outcome: Example –C4

Course outcome of the course “Applied Thermodynamics”

CO3:..... ?????

Example for CO3

- It is proposed that the exhaust gas from a natural gas-powered electric generation plant be used to generate steam in a heat exchanger. The exhaust gas, whose flow rate is 2 kg/s, enters the heat exchanger at 400°C and must leave at 215°C. To limit the pressure drop within the tubes, the tube gas velocity should not exceed 25 m/s. If saturated water at 11.7 bar is supplied to the shell side of the exchanger, choose an appropriate heat exchanger with high thermal efficiency. Assume that the properties of the exhaust gas can be approximated as those of atmospheric air and that the water-side thermal resistance is negligible. However, account for fouling on the gas side of the tubes and use a fouling resistance of 0.0015 m²·K/W

Course Outcome: Example –C4

Course outcome of the course “Applied Thermodynamics”

CO3: Analyse and choose the appropriate heat exchanger for a given application considering high thermal efficiency

Example for CO3

- It is proposed that the exhaust gas from a natural gas-powered electric generation plant be used to generate steam in a heat exchanger
.....However, account for fouling on the gas side of the tubes and use a fouling resistance of $0.0015 \text{ m}^2 \cdot \text{K/W}$

The Program Outcomes (POs)

- PO1 Apply Knowledge
- PO2 Problem Analysis
- PO3 Design/Development of Solution**
- PO4 Conduct Investigations
- PO5 Use Modern Tool
- PO6 Engineer and Society
- PO7 Environment and Sustainability
- PO8 Professional Ethics
- PO9 Individual and Team work
- PO10 Communicate Effectively
- PO11 Project Management and Finance
- PO12 Life-Long Learning

PO-3: Design/Development of Solutions

Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations

Course Outcome: Example -D1

At the end of the course the student will be able to **Design** filters to meet given specifications

Course Outcome: Example -D1 Assessment -1

Design and give the OP-AMP realization of the analog filter to have 3 dB cut-off at 1 KHz, and at least 12 dB attenuation at 10 KHz.

“Chebyshev or Butterworth”

Course Outcome: Example -D1 Assessment -1

Design and give the OP-AMP realization of the analog filter to have 3 dB cut-off at 1 KHz, and at least 12 dB attenuation at 10 KHz.

“IIR or FIR”

Course Outcome: Example -D1 Assessment -2

Design and arrive at the practical realization of the Digital filter to have 3 dB cut-off at 1 KHz, and at least 12 dB attenuation at 10 KHz, when the sampling frequency is 40 KHz.

Course Outcome: Example –D2

CO of the course “ADVANCED DESIGN OF REINFORCED CONCRETE STRUCTURES”

CO3: Design the reinforced cement concrete elements and detailing of structural members with safety, serviceability and ductility requirements as per codal provisions

Example for CO3

1. A reinforced concrete rectangular beam is supported on two walls 750 mm thick, spaced at a clear distance of 6 m. The beam carries a super imposed load of 30 kN/m. Assume suitable grade of concrete and reinforcements.

Course Outcome: Example –D3

Course Name	Course Outcome
Advanced Data Structures	?????????????????

Assessment:

Design and Implement a phone book management encompasses searching, sorting, and deleting operations. A distinctive feature of the search queries here is that the user sees suggestions from the contact list after entering each character.

Course Outcome: Example –D3

Course Name	Course Outcome
Advanced Data Structures	Design and Develop solutions for the various problems on non-linear data structures and pattern matching algorithms.

Assessment:

Design and Implement a phone book management encompasses searching, sorting, and deleting operations. A distinctive feature of the search queries here is that the user sees suggestions from the contact list after entering each character.

Course Outcome: Example –D4

Course outcome of the course “Thermodynamics”

CO4:..... ?????

Example for CO2

- Solar energy reaching the earth is about 1350 W/sq.m outside the earth's atmosphere, and 950 W/sq.m on earth's surface normal to the sun on a clear day. Someone is marketing 2 m X 3 m photovoltaic cell panels with the claim that a single panel can meet the electricity needs of a house. How do you evaluate this claim? Photovoltaic cells have a conversion efficiency of about 15 percent.

Course Outcome: Example -D1 to D3:

Observations

- ❖ No unique solution
- ❖ Many correct solutions
- ❖ Justify the selection and solution

The Program Outcomes (POs)

- PO1 Apply Knowledge
- PO2 Problem Analysis
- PO3 Design/Development of Solution
- PO4 Conduct Investigations**
- PO5 Use Modern Tool
- PO6 Engineer and Society
- PO7 Environment and Sustainability
- PO8 Professional Ethics
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PO-4: Conduct Investigations of Complex Problems

Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions

Course Outcome: Example –E

Course	Course Outcome
Analog Electronics	Ability to conduct experiments to validate few parameters from the data sheet of the identified analog electronic component
Analog Communication	Ability to implement the end-to-end analog communication system, on the Matlab/LabVIEW platform, and arrive at desired performance characteristics

Course Outcome: Example –E1

Course Name	Course Outcome
Web Application Development Using Java	Conduct experiments integrating Servlet and JSP with ORM for a given application / problem statement.

Assessment:

Develop simple features of User Management web application:
Create a User, Update a User, Delete a User, Retrieve a User, List of all Users

Course Outcome: Example –E2

Course outcome of “DYNAMICS OF MACHINERY”

CO4:.....????

Assessment:

1. Determination of natural frequency of torsional vibration of two rotor system.

Course Outcome: Example –E2

Course outcome of “DYNAMICS OF MACHINERY”

CO4: Conduct experiments on mass balancing and vibration.

Assessment:

1. Determination of natural frequency of torsional vibration of two rotor system.

Defining Outcomes for a Course

Course Outcomes: Observations

- COs are written for the skill to be addressed
- COs are NOT written for the topic
- COs are NOT abridged version of the course content
- Not advisable to have TWO COs to map to the same PO/PSO
- However, ONE CO can map to more than one PO/PSO
- COs are closely linked to the type of questions/ assignments/ Activities used for assessment
- A Course may have 4-6 COs and hence may map to 4 or more POs
- Not advisable to have ONE course map to ALL POs

Syllabus

EC3352 DIGITAL SYSTEMS DESIGN L T P C

3 0 2 4

COURSE OBJECTIVES:-

- To present the fundamentals of digital circuits and simplification methods
- To practice the design of various combinational digital circuits using logic gates
- To bring out the analysis and design procedures for synchronous and asynchronous Sequential circuits
- To learn integrated circuit families.
- To introduce semiconductor memories and related technology

UNIT I BASIC CONCEPTS

9

Review of number systems-representation-conversions, Review of Boolean algebra-theorems, sum of product and product of sum simplification, canonical forms min term and max term, Simplification of Boolean expressions-Karnaugh map, completely and incompletely specified functions, Implementation of Boolean expressions using universal gates, Tabulation methods.

UNIT II COMBINATIONAL LOGIC CIRCUITS

9

Problem formulation and design of combinational circuits - Code-Converters, Half and Full Adders, Binary Parallel Adder – Carry look ahead Adder, BCD Adder, Magnitude Comparator, Decoder, Encoder, Priority Encoder, Mux/Demux. Case study: Digital trans-receiver / 8 bit Arithmetic and logic unit, Parity Generator/Checker, Seven Segment display decoder

UNIT III SYNCHRONOUS SEQUENTIAL CIRCUITS

9

Latches, Flip flops – SR, JK, T, D, Master/Slave FF, Triggering of FF, Analysis and design of clocked sequential circuits – Design - Moore/Mealy models, state minimization, state assignment, lock-out condition circuit implementation - Counters, Ripple Counters, Ring Counters, Shift registers, Universal Shift Register. Model Development: Designing of rolling display/real time clock

UNIT IV ASYNCHRONOUS SEQUENTIAL CIRCUITS

9

Stable and Unstable states, output specifications, cycles and races, state reduction, race free assignments, Hazards, Essential Hazards, Fundamental and Pulse mode sequential circuits, Design of Hazard free circuits.

UNIT V LOGIC FAMILIES AND PROGRAMMABLE LOGIC DEVICES

9

Logic families- Propagation Delay, Fan - In and Fan - Out - Noise Margin - RTL, TTL, ECL, CMOS - Comparison of Logic families - Implementation of combinational logic/sequential logic design using standard ICs, PROM, PLA and PAL, basic memory, static ROM, PROM, EPROM, EEPROM EAPROM.

45 PERIODS

Course Outcomes (COs)

PRACTICAL EXERCISES:

30 PERIODS

1. Design of adders and subtractors & code converters.
2. Design of Multiplexers & Demultiplexers.
3. Design of Encoders and Decoders.
4. Design of Magnitude Comparators
5. Design and implementation of counters using flip-flops
6. Design and implementation of shift registers.

COURSE OUTCOMES :

At the end of the course the students will be able to

CO1: to **explain** the concepts of Digital Electronic circuits and Systems

CO2: to **apply** the concepts and simplification methods of Digital circuits for the given application

CO3: to **analyse** the given digital electronic circuits and arrive at suitable conclusions

CO4: to **design** a digital circuit for the given application and constraints

CO5: to **conduct experiments** to demonstrate the operation of digital electronics using suitable digital ICs

CO6: to **implement** the given digital circuit in 'Multisim' and perform analysis

TOTAL: 75 PERIODS

Mapping Strength

S.No	Course Outcome	PO(s) addressed	Contribution (%) for Assessment	Mapping Strength*
1	CO1: Remember and Understand	-	15 %	-
2	CO2: Apply	PO1	30 %	3
3	CO3: Analyse	PO2	25 %	3
4	CO4: Design and Development	PO3	10 %	1
5	CO5: Lab – Conduct Experiments	PO4, PO9	15 %	2
6	CO6: Lab – Simulate and Analyse	PO5, PO9	5 %	1

* Mapping Strength Guidelines

≥ 25 %	- 3
≥ 15 % and < 25 %	- 2
< 15 %	- 1

CO-PO Mapping

COs Vs POs & PSOs MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
1													2	2
2	3												3	2
3	3	3											3	2
4	3	2	1										3	2
5				2					2				2	2
6					1				1				3	2
@	3	3	1	2	1				2				3	2

@ - Overall Contribution to PO

Articulation Matrix for IV Semester (B.E – ECE)

Course	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
EC3452	3	2	1	--	--	2	2	--	--	--	--	--	1	3
EC3401	3	3	3	1	3	1	--	2	--	--	--	2	--	2
EC3451	3	2	2	--	--	1	--	--	2	--	2	--	3	--
EC3492	3	3	3	2	3	--	--	--	3	--	--	1	1	2
EC3491	3	2	--	--	2	--	--	--	--	--	--	--	2	3
EC3461	3	2	1	2	3	--	--	--	2	1	--	1	2	3
EC3462	3	2	1	--	1	--	--	--	2	1	--	1	3	

Program Articulation Matrix: B.E – ECE

Course	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
HS3152	1					2			2	3				
MA3151	3				1									
PH3151	3			2	1				2		2	1		
CY3151	2	1				2	2	1				1		
GE3151	3	2			1								1	
GE3171	3	2	1		1				3			1	1	
BS3171	3	3	3	1	1									
GE3172	3	2			1	1			3	3				
HS3252	1					2			2	3				
MA3251	3	2	1						1				1	
PH3254	3	3		2	1				2		2	1		

Program Articulation Matrix: B.E – ECE

Contd.,

Course	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
HS3152	2	3	3	3	2	3	3	3	2	3	3	3		
MA3151	3				1									
PH3151	3	3		2	1				2		2	1		
CY3151	2	1				2	2	1				1		
GE3151	3	2			1								1	
GE3171	3	2	1		1				3			1	1	
BS3171	3	3	3	1	1									
GE3172	2				1	1			1	3				
HS3252	1					2			3	3				
MA3251	3	2	1						1				1	
PH3254	3	3		2	1				2		2	1		
BE3254	3	2	1	3		1							2	1
GE3251	3	1	2		2					3		2		

Program Articulation Matrix: B.E – ECE

Contd.,

Course	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
EC3401	3	3	3	1	3	1		2				2		2
EC3451	3	2	2			1			2		2		3	
EC3492	3	3	3	2	3				3			1	1	2
EC3491	3	2			2								2	3
EC3461	3	2	1	2	3				2	1		1	2	3
EC3462	3	2	1		1				2	1		1	3	
EC3501	3	2		1	3				3			2		3
EC3552	3	3	3	2	2				2		1	3	3	1
EC3551	3	2	3											3
EC3561	3	3	3		3				2				3	
ET3491	3	3	3	3	3	2	2	2	3	2	3	2	3	1
CS3491	3	3	3	2	2	3		1					2	
CEC361	3	2			3								2	
CEC370	3	3	2	3	2		1		2		2		3	1
CEC362	3	2											2	
CEC366	3	3	2	2	2	1			2			2	2	
CEC352	3	2							3	3		1	2	
OEI354	3	2	2			1			3	2		2	2	
OMR351	3	3	3	3		1						3	3	2
EC3811	3	3	3	2	3	2	2	2	3	3	3	3	2	2